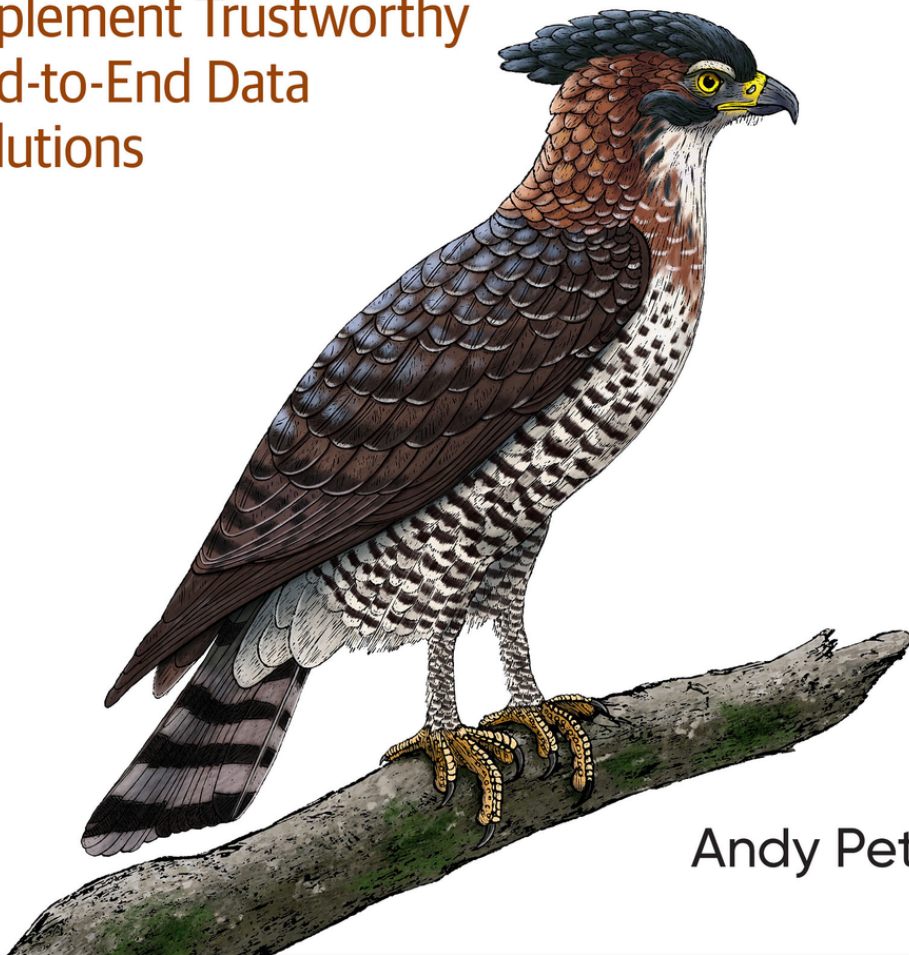


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Fundamentals of Data Observability

Implement Trustworthy
End-to-End Data
Solutions



Andy Petrella

Fundamentals of Data Observability

Many people talk about data observability, yet only some fully understand it. This is where this book comes in, drawing from Andy's professional background and vast hands-on experience, it offers a simple-to-implement, smart, and applicable structure for considering and applying data observability.

—Adi Polak, author of *Scaling Machine Learning with Spark*

Data observability is a topic that hasn't received the extensive discussion it deserves. Andy Petrella does an excellent job conveying the what, why, and how of achieving successful observability within data products. This will help guide the future of data-driven decision making.

—Matthew Weingarten, Data Engineer/Data Passionista

Data observability is widely discussed and widely misunderstood. In this book, Andy provides an intelligent and practical framework for thinking about and implementing data observability.

—Joe Reis, coauthor of *Fundamentals of Data Engineering* and “recovering data scientist”

This book is a brilliant manifestation of Andy's extensive experience in data engineering and architecture in a career that has spanned in-the-trenches pipeline construction, software development, product design, and leadership. He guides the reader through data observability on all levels, bridging the gap between engineering technical mastery and organizational dynamics. This is the book we've been waiting for to move beyond the hype and make data observability a reality.

—Matthew Housley, CTO and coauthor of *Fundamentals of Data Engineering*

I've seen businesses of all sizes struggle with data platform reliability time and time again. Andy's book provides both a theoretical and practical foundation for addressing this challenge head-on. The text explores the principles of data observability and backs these up with technical implementations designed to address the common and repeated patterns that plague data teams. This book is a must-read for any technologist tired of struggling with data quality, unreliable platforms, and opaque data pipelines.

—Emily Gorcenski, Principal Data Scientist, Thoughtworks

*Observability has gone mainstream in the past years, but one application area has been missing so far: data. With *Fundamentals of Data Observability*, you now have the missing piece at hand, explaining what observability means for your data pipelines and how to successfully implement it in your organization. An excellent hands-on guide that should be at the top of your reading list!*

—Michael Hausenblas, AWS

This book paints a realistic and accurate view of data observability, shares the key tenants and mental models, and proves that observability is a critical requirement for any modern data organization and partitioner looking to remain competitive.

—Scott Haines, Distinguished Software Engineer, Nike

Fundamentals of Data Observability

Implement Trustworthy End-to-End
Data Solutions

Andy Petrella

Fundamentals of Data Observability

by Andy Petrella

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Preface

Welcome to *Fundamentals of Data Observability*, a book designed to provide a robust introduction to a crucial, emerging field in data engineering and analytics.

As we venture into an era characterized by unprecedented data growth, the importance of understanding our data—its sources, destinations, usages, and behaviors—has never been more important. Observability, traditionally a term associated with software and systems engineering, has now made its way into the data space, becoming a cornerstone of trustworthy, efficient, and insightful data systems. This book aims to guide readers into the depth of this new and necessary discipline, exploring its principles, techniques, and evolving best practices.

Fundamentals of Data Observability is not just for data engineers or data scientists, but for anyone who interacts with data systems in their daily work life. Whether you're a chief data officer (CDO), a chief technology officer (CTO), a manager, a leader, a developer, a data analyst, or a business manager, understanding data observability concepts and principles will empower you to make better decisions, build more robust systems, and gain greater insight from your data resources.

This book begins by outlining the core concepts of data observability, drawing parallels to similar concepts in software engineering, and setting the stage for the more advanced material. It subsequently delves into the principles and techniques to achieve data observability, providing practical guidance on how to implement them. The final section discusses how to get started today with the system you are

using or have inherited. The book concludes with thoughts about the future of data observability, exploring ongoing research and emerging trends that are set to shape the field in the coming years.

Every chapter in this book is packed with actionable advice to reinforce the topics covered. My aim is not merely to impart knowledge but to facilitate the practical application of data observability concepts in your real-world situations.

I hope that by the end of this book, you not only will understand the “what” and “why” of data observability but will also be armed with the “how”—practical knowledge that you can apply to improve the reliability, usability, and understandability of your data systems.

The field of data observability is still young, and there is much to explore and learn. As you embark on this exciting journey, remember that understanding our data and its usages is not just a technical goal—it’s a foundation for making better decisions, fostering innovation, and driving the success of our enterprises.

Overview of the Book

Fundamentals of Data Observability is organized into three parts and eight chapters, each addressing specific areas of data observability:

Part I: The Foundations of Data Observability

Chapter 1 introduces the concept of data observability, explains why it has become an essential aspect of data management and sheds light on its role in facilitating accurate, reliable, and actionable insights from data.

Chapter 2 delves deeper into the components of data observability. It provides an understanding of its multifaceted nature and how to ensure its implementation aligns with both short-term and long-term organizational needs.

Chapter 3 describes the roles and responsibilities of a data observability platform within an organization. It covers its relationships with existing systems and practices, and explores the changes in workflow and management that it facilitates.

Part II: Implementing Data Observability

Chapter 4 outlines the APIs and practices required to make a data system observable. It includes a Python-based implementation example that can be adapted to suit any other programming language or system.

Chapter 5 focuses on automating the generation of data observations. It showcases a range of practices to lessen manual effort, thereby enabling data users to concentrate on more strategic tasks.

Chapter 6 covers how to implement expectations within data applications, supporting continuous validation, one of the crucial principles of data observability.

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Part III: Get Started Today

Chapter 7 provides ready-to-use recipes for implementing data observability principles into various technologies used in data pipelines. The applications range from traditional data processing systems to machine learning applications and data visualization tools, broadening the scope of data observability.

Chapter 8 provides actionable intelligence to incorporate data observability into systems that are currently opaque or closed, or whose knowledge base has faded over time.

Through these three sections, *Fundamentals of Data Observability* offers a comprehensive guide to understanding, implementing, and leveraging data observability principles in various data systems, new or existing.

Who Should Read This Book

Fundamentals of Data Observability is a vital guide for anyone who plays a role in the world of data engineering, analytics, and governance. This book provides in-depth insight into the principles of data observability and its role in ensuring efficient and reliable data systems. Here's who should read this book, and why:

Data engineers and analyst engineers

Whether you are just beginning your career or are an experienced professional, the book will empower you with the knowledge to architect and manage observable data systems. It delves deep into the tools, technologies, and practices that can improve the reliability, usability, and understandability of data systems. The book also provides practical advice and case studies to help you apply these concepts in real-world scenarios.

Lead data engineers and heads of data engineering

This book is a resource for team leaders and managers responsible for designing, implementing, and managing data systems. The chapters guide you on creating a strategy for implementing data observability in your organization and offer advice on managing the change effectively. It provides the knowledge you need to mentor your team and facilitate their growth in this emerging field.

CDOs, CTOs, and heads of data

For those in executive roles, the book offers an overview of the principles of data observability and its significance in the broader data architecture landscape. Understanding this will allow you to make informed decisions about resource allocation, risk management, and strategic direction. It provides a firm grounding in the language and concepts of data observability, enabling you to engage more productively with your technical teams.

Data governance and architecture professionals

For those involved in data governance and architecture, this book provides insights into how data observability principles can contribute to robust, secure, and compliant data practices. It addresses how data observability intersects with other data systems in place, helping you build a more integrated and effective data strategy.

Software engineers

If you are a software engineer working on building data systems, the book's second part will be especially relevant to you. It provides practical guidance on how to make these systems observable, thereby ensuring they can be effectively maintained and their data properly utilized by data users such as engineers and analysts.

In a world increasingly dominated by data, understanding the principles of data observability is crucial. This book will equip you with the knowledge and skills to make your data systems more reliable, understandable, and usable, driving better decision making and business success. Whether you are a hands-on engineer, a team leader, or a strategic decision maker, *Fundamentals of Data Observability* is an essential addition to your professional library.

Conventions Used in This Book

The following typographical conventions are used in this book:

Italic

Indicates new terms, URLs, email addresses, filenames, and file extensions.

`Constant width`

Used for program listings, as well as within paragraphs to refer to program elements such as variable or function names, databases, data types, environment variables, statements, and keywords.

TIP

This element signifies a tip or suggestion.

NOTE

This element signifies a general note.

WARNING

This element indicates a warning or caution.

Using Code Examples

Supplemental material (code examples, exercises, etc.) is available for download at <https://github.com/Fundamentals-of-Data-Observability/oreilly-fodo-source-code>.

If you have a technical question or a problem using the code examples, please send email to support@oreilly.com.

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To everyone who contributed, your collective efforts have shaped this book into what it is today. My sincerest thanks to you all.



Part I. Introducing Data Observability

Your File Format APIs

Chapter 1. Introducing Data Observability

Once upon a time, there was a young data analyst named Alex who had a deep passion for data. Alex loved the way data could help businesses make informed decisions, drive growth, and achieve success. However, Alex was also aware of the dangers of misinterpreting data or not having enough visibility into the data.

Alex was working on a critical project with a data engineer named Sarah. Sarah was responsible for preparing the data and making sure it was ready for analysis. As they delved deeper into the project, Alex and Sarah realized that there were many variables at play, and the data they were working with was not as straightforward as they had initially thought.

One day, while Alex was iterating on his analysis to generate insights, it appeared to him that the results presented on that day were looking odd and hard to relate to what he had seen to that point. He went to Sarah to discuss the case, but Sarah needed more context on his previous interpretation was or what his expectations were, and what he was asking her to check.

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After four days of collaboration, pair review, and several brainstorming sessions, Sarah and Alex discovered that subtle changes in the distribution of half a dozen variables of the incoming data shifted the generated insights, several transformation steps later. Some variables had more missing values, hence they were dropped in the cleaning transformation, others had their average value increased greatly, and one of the datasets was refreshed with almost twice as much data as it had in the past thanks to better extraction from the operational sources.

Although Sarah and Alex originally thought the quality of the data may have dropped, it appeared that the data simply changed, and their assumptions about the data had to be aligned. They realized that they were lucky this had happened before deploying the project in production, and this kind of situation would probably happen many times in the future. If they didn't anticipate these changes, they would be in trouble.

That experience made them realize the importance of data observability. They needed to have visibility into the data, its transformations, and its usage so they would be able to react quickly to any changes. They started to embrace data observability principles and instrumented their data pipelines to add the needed capabilities that provide real-time insights into the data, its quality, and its usage.

From that day on, they have been backed with dashboard and notification systems that track the health of the data in the pipelines and alert them to any issues that need attention to ensure that the client always receives accurate and reliable data.

Through this experience, Alex and Sarah learned that data is a living and breathing entity that needs to be continuously monitored and observed. They realized that, without data observability, they never would be able to react quickly to changes, putting the project's success at risk.

If you're reading this book, it's likely because you, like Sarah and Alex, have experienced or anticipate experiencing similar situations in your own work with data. You understand the power of data, but also the potential for disastrous consequences when even subtle changes go unnoticed.

But you don't know what you need to do or how to get started. There's no need to worry anymore. In this book you'll learn what Alex and Sarah did to embrace data observability principles and ensure that their data pipelines are reliable and that their clients receive accurate and trustworthy data. By applying these principles to your own work, you can avoid the pitfalls of unreliable data and build a strong foundation for success in your data-driven projects.

Before I dig further into what data observability is and offers at scale, let's first look at how data teams are evolving and identify the challenges they face.

Scaling Data Teams

More roles, more responsibility, more engineering.

A data team is a group of data practitioners who work together to collect, process, analyze, and interpret data to generate insights and inform decision-making to drive better business outcomes and improve organizational performance.

Because data teams play a strategic role in organizations, their operational efficiency can become a bottleneck to incorporating high-demand data into critical business operations. To cope with this, data teams evolved similarly to IT teams in the 1950s—dedicated roles such as systems engineers, web engineers, or backend engineers were added to support specific operations instead of having generalist roles.¹

As data teams have increased their operational complexity with additional roles, more interactions and interdependencies across members and teams have been created, increasing the need for greater visibility, transparency, and standardization.

In the remainder of this section, I will use search trends as an example that illustrates the creation and scaling of data teams along with the data engineer role. The impact and challenges coming with it are addressed in the following section, which also describes other roles, such as analytics engineers, for example.

Let's start with a Google Trends search (Figure 1-1) for the term “data team” from 2004 to 2020. We can see that while data teams have always been around, the interest in data teams began increasing in 2014 and accelerated significantly between 2014 and 2020.

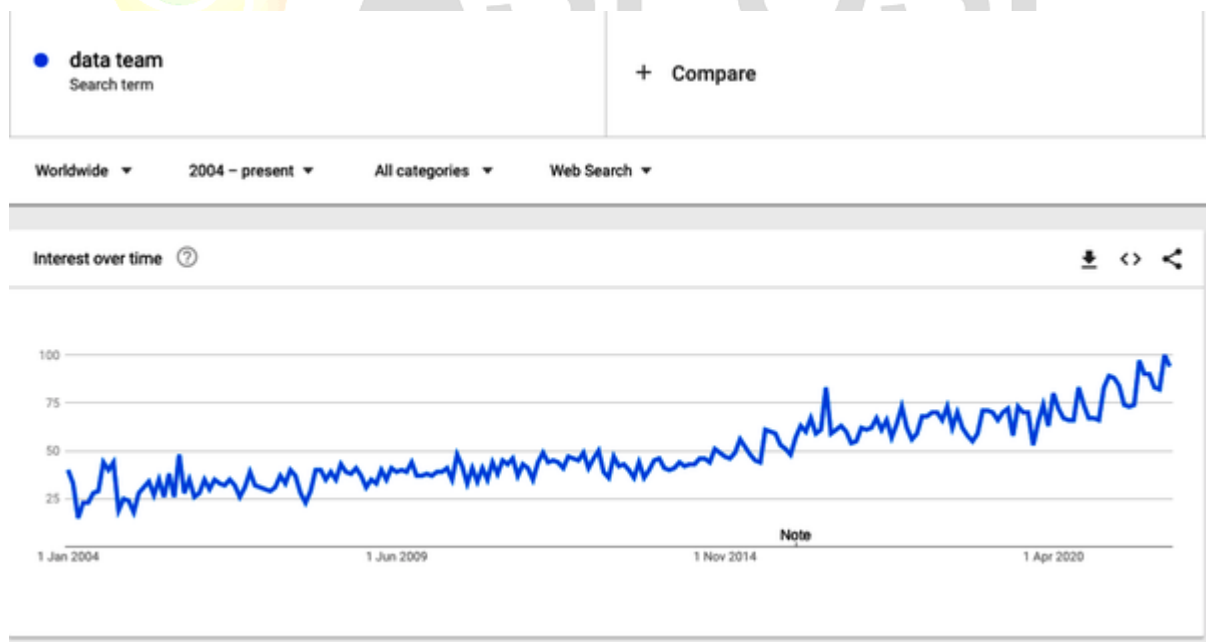


Figure 1-1. Google search trend for “data team”

This is despite interest in “Big Data” decreasing (as shown in Figure 1-2).

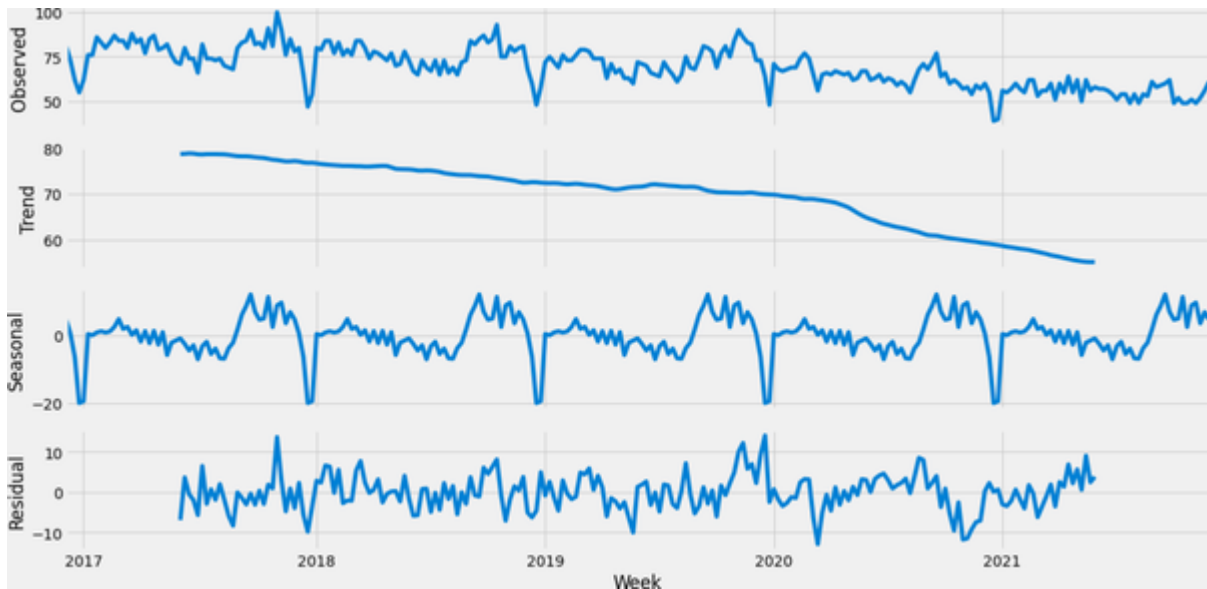


Figure 1-2. Analysis of “Big Data” search term on Google Trends²

Obviously, this doesn’t mean that Big Data was not needed anymore, but, as Figure 1-3 shows, in 2014 the focus started to shift toward data science because its link with value generation was more intuitive.

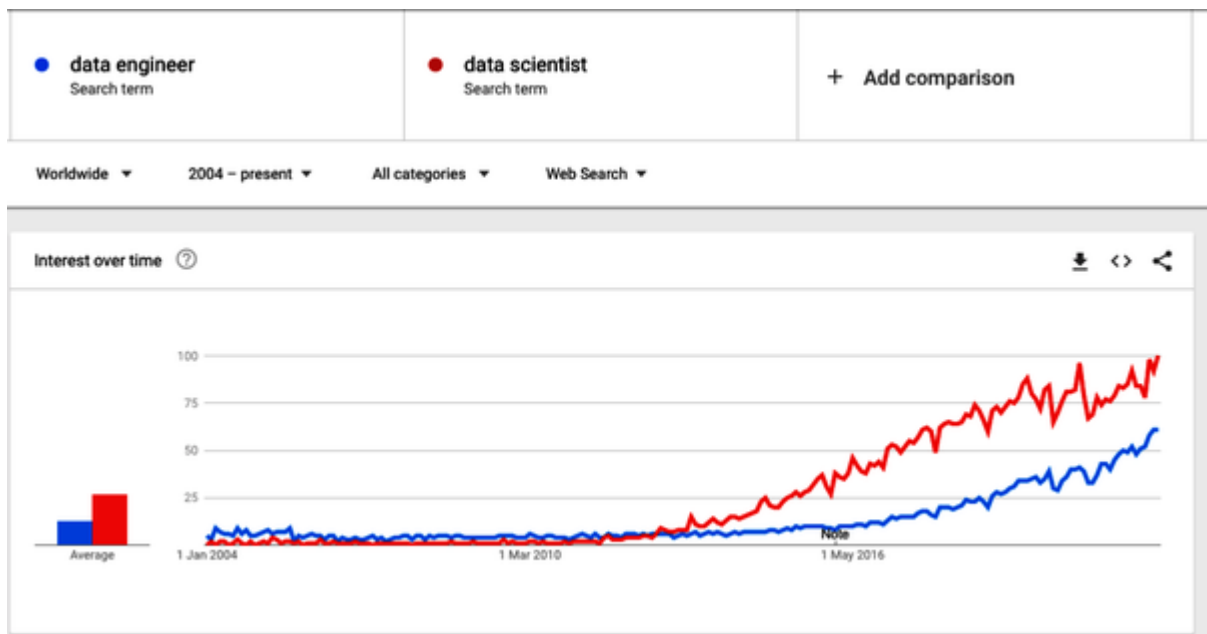


Figure 1-3. Google trends for search terms “data engineer” and “data scientist”

However, even as interest began to climb in data science, it was clear that data availability was a roadblock to many data science projects.

Data science teams have not replaced big data teams. Instead, data teams have embraced analytics, and thus added this relatively new role of data scientist.

As data and analytics became more central to companies’ success, providing the right data to the right people has remained a constant challenge. Gartner noted that by 2018,³ data engineers had become

crucial in addressing data accessibility challenges and that data and analytic leaders must, therefore, “develop a data engineering discipline as part of their data management strategy.”

Hence, it became evident that a role dedicated to producing data for downstream use cases was missing. This is why, since around 2020, as companies began to bring on engineers specifically to help build data pipelines and bring together data from different source systems, the search volume for data engineers has increased significantly. Today, as also shown in **Figure 1-3**, data engineering is trending toward catching up to data scientist in search popularity.

DATA AS A METRIC

Data availability is one of the metrics driving data observability. There are important metrics and metadata relating to data availability, such as the time to live (TTL) and refresh rate (freshness) of a dataset. Still, there are also nuances we’ll get into throughout the book such as sourcing data, privacy and security ramifications, etc. Meanwhile, though, separating roles allows the different skills time and space to be built and the resources to be focused on the different phases of data projects.

As previously introduced, scaling data teams this way with more people, roles, and responsibilities comes with its own challenges. Let’s dig into that topic in the next section.

Challenges of Scaling Data Teams

As companies look to scale their data usage, they also must scale their data teams. I will cover the challenges and their consequences with the following examples:

- A description of the emergence of roles and responsibilities in a data team.
- An analysis of the operational complexity as the team grows.
- An illustration of the challenges of (data) issuance management and their impact on the mood and efficiency of the data team.
- A note of caution to avoid failures on the ML/AI journey.

Consider a data team that starts with a single person tasked to do the data ingestion, data integration, and even the final report. For this person, a first project might consist of producing a view of recent customer acquisitions.

The data engineer would have to query and ingest the source data from the company’s *customer relationship management* (CRM) system into the data lake, integrate it into the data warehouse, and create a report that gives the head of sales insight into the progress of the sales made over the past few weeks.

This first data team member is, in essence, a Swiss Army knife: covering both data engineering and data analysis responsibilities to deliver the project outcome (report).

In a larger data team, the skills required for this work are balanced between data engineering, data analysis, and business analysis. As a team of one, though, an individual must master the technologies required to build and run the pipeline, and reporting tools, and understand business key performance indicators (KPI)s. It is worth noting that each of these areas requires specific expertise.

So far, this person is happy, and the scope of work and responsibilities are as shown in **Figure 1-4**.

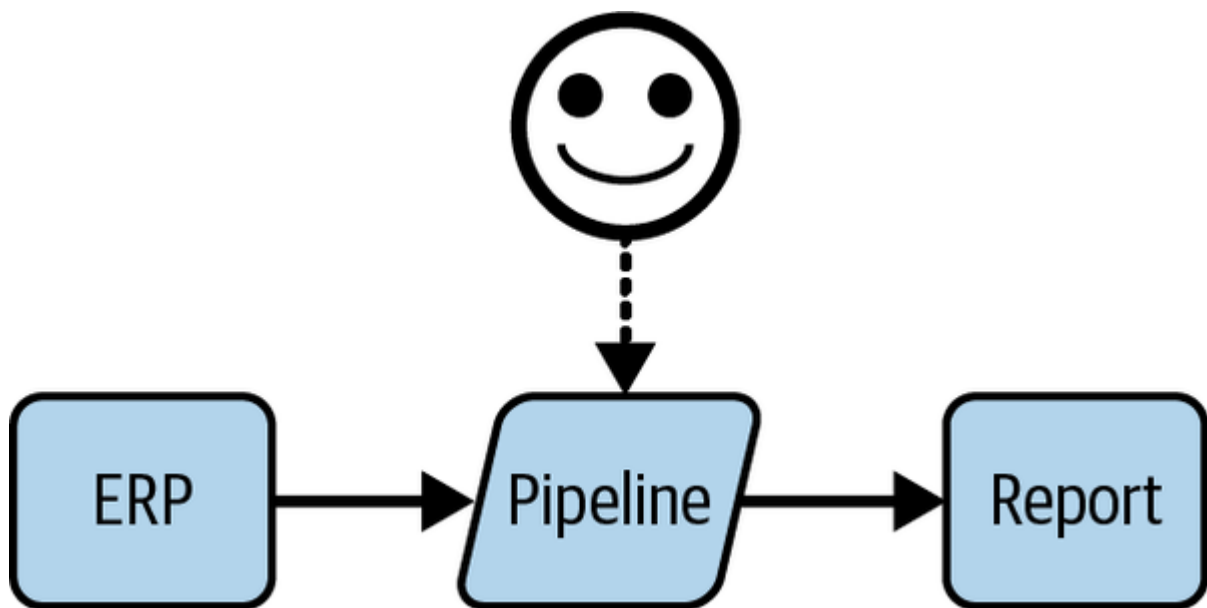


Figure 1-4. A happy single-person data team and its work

Of course, the process is not quite as simple. For example, an entire systems development life cycle (SDLC) should be followed to get the data into production and allow it to be available for analysis by the final user. At a very abstract level, however, this basic process outlined is what we mostly think of as “producing value” with a data project.

With only one team member and one data project, the whole data ecosystem is relatively under control.

The process is simple enough for one person, and the responsibility for the entire project is also attributed to that individual. Because there is just a single person on the team, they also have a close relationship with the business stakeholders—such as the head of sales—and have, therefore, built some domain knowledge around the use case and the business objective of the project. While this role is typically what a business analyst would perform, in this case, the data engineer takes on this role as well.

Thus, if the user raises any issues, it is clear who should work on troubleshooting. After all, they not only executed each step of the process but also know each area of the process.

However, as stakeholders become aware of the potential and value data can provide, more requests are made to this person for new reports, information, projects, etc. As a result, there is an increase in the

complexity of the demands and ultimately, the individual reaches their limits of production capacity under the current setup. Consequently, it becomes urgent to grow the team and invest in specialized members to optimize productivity.

In the next section, we'll see how this team starts growing as specific roles are created, as presented in **Table 1-1**, which also highlights each role's responsibilities and dependencies.

Table 1-1. Data team (extended) roles, responsibilities, and dependencies.^a

Title/role	Responsibilities	Dependencies
Software and Operation Engineer	Build and maintain the platform; manage security and compliance; oversee site reliability engineering (SRE)	Receive recommendations from other teams about what to monitor and what to consider an error
Data Engineer	Build the data pipelines and manage the data from deployment to production	Rely on IT to build the platform and set up the tools to monitor the pipeline and systems, including data observability tools
Analytics Engineer	Similar to data engineering with a stronger focus on data modeling and reusability	Rely on data engineering team to make the operational data sources available in the analytic platform
Analytic/data scientist	Build analytics and AI/ML models that can analyze and interpret data for the business team's use cases	Rely on the data and analytic engineering team to build the pipeline to generate the data they will be using in their report and machine learning models

This document was truncated here because it was created in the Evaluation Mode.